

Oru'el GPU API Reference

By integrating with this API, you get direct programmatic access to rent, manage, and scale GPU instances.

1. Getting Started

Base URL

All API requests must be made to the following base endpoint:

```
https://relay.oru-el.com/api
```

Authentication

To authenticate requests, you must supply the API key issued to you by Oru'el in the `X-API-Key` HTTP header.

- Do not share this API key.
- Never hardcode the API key in client-side applications.

Header Format:

```
X-API-Key: oruel_live_XXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

cURL Authentication Example:

```
curl -H "X-API-Key: oruel_live_YOUR_API_KEY_HERE" \  
https://relay.oru-el.com/api/providers
```

2. Pricing & Markup

All currency values (lowestPrice, price, hourlyRate, totalCost, etc.) returned by the Oru'el API are in **USD** and **already include markup**.

There is no need to manually calculate markups on your end; the rate you see is the exact billing rate you accrue.

3. Rate Limits (Can be modified as per demand)

Throttling is applied per-client (associated with your API key) to protect upstream providers:

- **General Endpoints:** Maximum of `250 requests` per 15-minute window.
- **Deployment Creation (`POST /deployments`):** Maximum of `10 requests` per 15-minute window.

If you exceed these limits, the API returns a `429 Too Many Requests` status code.

4. Endpoints Reference

GPU Offers

`GET /gpu-offers`

Retrieve the real-time paginated catalog of available GPU configurations across all providers.

- **Query Parameters:**
 - `page` (integer, default: `1`): Page number to fetch.
 - `limit` (integer, default: `10`): Number of GPU model groups per page.
 - `search` (string, optional): Filter by GPU name (e.g. `rtx-4090` , `h100`).
 - `sortBy` (string, default: `popularity`): Field to sort by.
 - `sortOrder` (string, default: `asc`): Sort order (`asc` or `desc`).
 - `instanceType` (string, optional): filter by `SPOT` , `DEDICATED` , or `CLUSTER` .
- **Response Example (200 OK):**

```
{
  "data": [
    {
      "gpuType": "rtx-4090",
      "gpuModel": "RTX 4090",
      "displayName": "NVIDIA GeForce RTX 4090",
      "popularity": 95,
      "totalAvailable": 12,
      "lowestPrice": 0.588,
      "highestPrice": 0.828,
      "averagePrice": 0.708,
      "providers": ["massed-compute", "sesterce"],
      "offers": [
        {
          "provider": "massed-compute"
```

```

        "provider": "massed-compute",
        "offerId": "offer-mc-4090-single",
        "name": "Single RTX 4090 Dedicated",
        "description": "NVIDIA RTX 4090 with Ubuntu 22.04 LTS",
        "vcpus": 8,
        "memory": 32,
        "storage": 100,
        "gpuCount": 1,
        "price": 0.588,
        "available": true,
        "region": "us-east",
        "gpu_memory": 24,
        "os_options": ["ubuntu-22.04"],
        "instanceType": "DEDICATED",
        "interconnectType": "PCIe",
        "supportsCloudInit": true
    }
]
}
],
"total": 1,
"page": 1,
"limit": 10,
"totalPages": 1
}

```

Deployments

POST /deployments

Launches a new GPU instance or Kubernetes cluster.

- **Request Body (JSON):**

- `provider` (string, required): Upstream provider ID (e.g. `massed-compute`).
- `offerId` (string, required): The target `offerId` from `/gpu-offers`.
- `gpuType` (string, required): Hardware type (e.g. `rtx-4090`).
- `gpuCount` (integer, required): Number of GPUs to rent.
- `region` (string, required): Deployment region identifier.
- `operatingSystem` (string, required): OS image identifier.
- `instanceType` (string, required): `SPOT`, `DEDICATED`, or `CLUSTER`.
- `sshKeyId` (string, optional): ID of a previously uploaded SSH key.
- `ssh_public_key` (string, optional): Literal SSH public key text to inject.
- `name` (string, optional): Custom friendly name for your deployment.
- `cloudInit` (object, optional): Custom startup script config (`runCmd`, `packages`, `writeFiles`).

- `volumeIds` (array of strings, optional): Volumes to attach at runtime.

- **Request Example:**

```
curl -X POST -H "X-API-Key: $ORUEL_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "provider": "massed-compute",
  "offerId": "offer-mc-4090-single",
  "gpuType": "rtx-4090",
  "gpuCount": 1,
  "region": "us-east",
  "operatingSystem": "ubuntu-22.04",
  "instanceType": "DEDICATED",
  "ssh_public_key": "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQBAQC..."
}' \
https://relay.oru-el.com/api/deployments
```

- **Response Example (201 Created):**

```
{
  "id": "dep-9a3b8c2d1e",
  "name": "instance-rtx-4090",
  "providerId": "massed-compute",
  "gpuType": "rtx-4090",
  "offerId": "offer-mc-4090-single",
  "gpuCount": 1,
  "region": "us-east",
  "status": "deploying",
  "hourlyRate": 0.588,
  "totalCost": 0.00,
  "vcpus": 8,
  "memory": 32,
  "storage": 100,
  "createdAt": "2026-07-20T10:00:00Z"
}
```

GET /deployments

List all running and terminated GPU deployments launched with **your** API key.

- **Query Parameters:**

- `status` (string, optional): Filter by `active`, `inactive`, `running`, `deploying`, `terminated`, `failed`.

GET /deployments/{id}

Returns real-time configuration, status, IP address, and connection details for a specific

deployment.

- **Response Example (200 OK):**

```
{
  "id": "dep-9a3b8c2d1e",
  "name": "instance-rtx-4090",
  "providerId": "massed-compute",
  "gpuType": "rtx-4090",
  "offerId": "offer-mc-4090-single",
  "gpuCount": 1,
  "region": "us-east",
  "status": "running",
  "ipAddress": "198.51.100.42",
  "sshCommand": "ssh root@198.51.100.42 -p 22",
  "sshPort": 22,
  "hourlyRate": 0.588,
  "totalCost": 1.176,
  "createdAt": "2026-07-20T10:00:00Z"
}
```

PATCH /deployments/{id}

Update mutable settings of a deployment.

- **Request Body:**

- `name` (string): The new name of the instance.

DELETE /deployments/{id}

Terminates a deployment immediately and stops billing accrual.

- **Response Example (200 OK):**

```
{
  "message": "Instance destruction initiated",
  "deployment": {
    "id": "dep-9a3b8c2d1e",
    "status": "terminated",
    "stoppedAt": "2026-07-20T12:00:00Z"
  }
}
```

GET /deployments/{id}/can-terminate

Check if the instance can be terminated (certain bare-metal providers enforce minimum runtime constraints).

- **Response Example (200 OK):**

```
{
```

```
"canTerminate": true,  
"runtime": 120,  
"minimumRuntime": 60,  
"timeRemaining": 0  
}
```

SSH Keys

Manage public keys utilized to secure and log in to newly provisioned deployments.

GET /ssh-keys

List SSH keys added to your key workspace.

POST /ssh-keys

Add an SSH key.

- **Request Body (JSON):**

- `name` (string, required): Friendly identifier for the key.
- `publicKey` (string, required): OpenSSH format public key content.

DELETE /ssh-keys/{id}

Remove a public SSH key from your workspace.

Volumes (Persistent Storage)

Renting external block storage and managing attachments.

GET /volumes

List storage volumes.

- **Query Parameters:**

- `page` (integer)
- `limit` (integer)
- `status` (string, optional): `available`, `attached`, `deleting`.

POST /volumes

Create a persistent storage volume.

- **Request Body (JSON):**

- `name` (string, required): Lowercase alphanumeric name with hyphens

- `name` (string, required): Lowercase alphanumeric name with hyphens.
- `sizeInGb` (integer, required): Volume capacity in GB.
- `provider` (string, required): Storage provider ID (e.g. `voltage-park`).
- `region` (string, required): Target region.

DELETE /volumes/{volumeId}

Deletes a volume (must be detached first).

POST /volumes/{volumeId}/attach

Attach volume to a running deployment.

• Request Body (JSON):

- `deploymentId` (string, required): Target instance deployment ID.

POST /volumes/{volumeId}/detach

Detach volume from an instance.

GET /volumes/pricing

Get storage rates per GB per hour (including Oru'el markup).

5. Error Reference

Standard HTTP response status codes are returned to indicate the success or failure of an API request:

Code	Type	Meaning
200	Success	Request succeeded.
201	Success	Resource created successfully.
400	Bad Request	Missing or invalid parameters in body/query.
401	Unauthorized	Missing or invalid <code>X-API-Key</code> header.
404	Not Found	Target resource does not exist or belongs to another client.
429	Rate Limited	Too many requests in a short period.
500	Server Error	Internal gateway or upstream error.

• Error Response Body Shape:

```
{
```

```
"error": "Brief description of the error",
"code": "ERROR_CODE",
"details": {}
}
```

6. End-to-End Workflow Example

The following scenario shows how to search for resources, create an SSH key, provision a GPU instance, and terminate it.

Step 1: Browse for RTX 4090 Offers

```
curl -H "X-API-Key: YOUR_API_KEY" \
  "https://relay.oru-el.com/api/gpu-offers?search=rtx-4090&instanceType=DEDICATED"
```

Record the chosen `offerId`, `provider`, and `region` from the response.

Step 2: Upload Your Public SSH Key

```
curl -X POST -H "X-API-Key: YOUR_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "name": "dev-laptop",
    "publicKey": "ssh-rsa AAAAB3Nza..."
  }' \
  https://relay.oru-el.com/api/ssh-keys
```

Note the returned `"id"` (e.g., `key-12345`) representing the uploaded key.

Step 3: Launch the GPU Instance

```
curl -X POST -H "X-API-Key: YOUR_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "provider": "massed-compute",
    "offerId": "offer-mc-4090-single",
    "gpuType": "rtx-4090",
    "gpuCount": 1,
    "region": "us-east",
    "operatingSystem": "ubuntu-22.04",
    "instanceType": "DEDICATED",
    "sshKeyId": "key-12345"
  }' \
  https://relay.oru-el.com/api/deployments
```


Record the deployment `"id"` (e.g. `dep-abcde`). The initial status will be `"deploying"`.

Step 4: Poll Instance Status Until Running

```
curl -H "X-API-Key: YOUR_API_KEY" \  
https://relay.oru-el.com/api/deployments/dep-abcde
```

Once the status changes to `"running"`, copy the `"ipAddress"` or `"sshCommand"` to access your VM.

Step 5: Terminate Instance (Stop Billing)

```
curl -X DELETE -H "X-API-Key: YOUR_API_KEY" \  
https://relay.oru-el.com/api/deployments/dep-abcde
```

Your instance is deleted and billing terminates instantly